Process Characterization Report

A-Mab Drug Substance — Harvest and Clarification (Step 4) — PCR-004

Process Development / Manufacturing Science & Technology (MSAT)

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Table of contents

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Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
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Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

AMV, analytical method validation; BLA, Biologics License Application; CPP, critical process parameter; CQA, critical quality attribute; DoE, design of experiments; DS, drug substance; GPP, general process parameter; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; KPP, key process parameter; LOQ, limit of quantitation; MSAT, Manufacturing Science and Technology; MVM, minute virus of mice; NOR, normal operating range; NTU, nephelometric turbidity unit; PAR, proven acceptable range; PPQ, process performance qualification; qPCR, quantitative polymerase chain reaction; rcf, relative centrifugal field; RPN, risk priority number; RSM, response surface model; SDM, scale-down model; SEC, size exclusion chromatography; SOP, standard operating procedure; WC-CPP, well controlled critical process parameter; XMuLV, xenotropic murine leukaemia virus.

Executive summary

Harvest and clarification is the primary recovery step of the A-Mab drug substance process. Whole cell broth from the production bioreactor is separated by continuous disk-stack centrifugation, and the centrate is then clarified through a depth filter train and a sterile filter (SOP-2005 and SOP-2006). The step delivers the clarified harvest that loads Protein A capture. It was characterized on a qualified scale-down model (SOP-1001) under the approved protocol PCP-004, within the lifecycle approach to process validation (U.S. Food and Drug Administration 2011) and the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012), and the scope of the study was fixed beforehand by the pre-characterization risk assessment RA-001.

The step forms no critical quality attribute. None of the 10 attributes in the drug substance register is assigned to harvest and clarification, and none of the 7 attributes present in the harvested stream is modified across it. The parameters of the step therefore act on process performance, and they were characterized against clarification efficiency and product yield.

The study covered 3 parameters. Each was assessed univariately over a characterization range between 2 and 3 times the width of its normal operating range (NOR), following the study assignment that RA-001 made for this step, and none of the three moved a quality attribute anywhere it was tested. The outcome was 2 key process parameters (KPP) and 1 general process parameter (GPP). No parameter of this step is a critical process parameter or a well controlled critical process parameter, because none of them is linked to a quality attribute. The classifications and the ranges that support them are given in Table 4.1.

The nominal batch recovered 97.7% of the product mass entering the step, at a post-clarification turbidity of 5 NTU, which leaves cumulative product yield at 97.7% against 83.2% for the whole drug substance train.

No commercial-scale process capability is attributed to this step, because it sets no attribute against which a capability could be computed. The impurity burden it delivers to capture does have a capability at drug substance, and that capability is met with wide margin. Host cell protein is the tightest of the three attributes carried through the step, at a capability index (Cpk) of 6.14 against an upper limit of 100 ng/mg, and that result is delivered by the three chromatographic clearance steps reported in PCR-005, PCR-007 and PCR-008 and not by harvest. No viral clearance is claimed for this step, and none is credited to it in the modular clearance assessment (International Council for Harmonisation 2023a).

Execution recorded 2 deviations (Section 13). Both were investigated, both were dispositioned as retained, and neither changed a parameter classification or a reported range. The step is well understood over the ranges studied, those ranges are justified for Stage 2, and every conclusion here is bounded by them, by the qualified scale-down model, and by a feed drawn from a culture operated inside the endpoint ranges characterized in PCR-003. This report rolls up into the master report PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody whose principal mechanism of action is antibody dependent cellular cytotoxicity. It is produced by fed-batch cell culture and purified by a platform train of Protein A capture, low-pH viral inactivation, cation exchange, anion exchange, small-virus retentive filtration and ultrafiltration with diafiltration. The train is shown in Figure 1.1. Harvest and clarification is Step 4, between the production bioreactor and Protein A capture.

The step operates in two stages. Whole cell broth is fed continuously to a disk-stack centrifuge, which sediments cells and the larger cell debris under an applied relative centrifugal field (rcf), and the centrate is then passed through a depth filter train, which retains the finer colloidal material that the centrifuge does not remove, and through a sterile filter before hold. Both stages are physical separations. Neither introduces a chemical condition capable of altering the antibody (no pH shift, no elution buffer, no hold at a denaturing condition), and neither discriminates between product variants.

The step forms no quality attribute of its own, so there is nothing here for a design space to protect. It also removes no soluble impurity, so the host cell protein (HCP) and DNA burden generated during culture passes forward essentially intact into the clarified harvest, and the acceptable output levels of both are set by what the downstream steps can clear. What the step controls is the particulate content of that stream and the product mass recovered with it. Post-clarification turbidity is the in-process measure of the first, and step yield is the measure of the second.

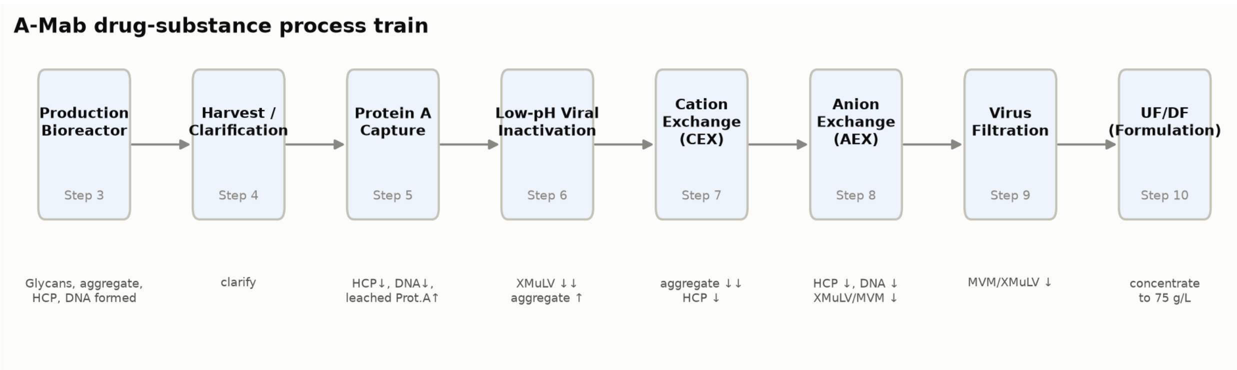

Figure 1.1: A-Mab drug substance process train with the principal quality impact of each unit operation.

1.2 Objectives

The objectives of the characterization study were the following.

1. To quantify the relationship between each harvest and clarification parameter and the performance of the step, over ranges wider than routine operation.
2. To confirm that no product quality attribute of A-Mab is altered across the step, and to record the evidence on which that confirmation rests.
3. To establish a proven acceptable range for each parameter and to locate the normal operating range inside it.
4. To classify each parameter under SOP-4001, from the effects observed in the study and from the risk of operating outside the characterized region.
5. To justify the ranges and the in-process controls that this step contributes to the Stage 2 control strategy and to the licence application.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the sense of the FDA lifecycle guidance, and this report is the study record for that stage (U.S. Food and Drug Administration 2011). The development approach is the enhanced approach of ICH Q8, applied through the quality risk management framework of ICH Q9 and the development and manufacture principles of ICH Q11 (International Council for Harmonisation 2009, 2023b, 2012). Criticality is treated as a continuum and not as a binary label, which is the position ICH Q9 takes and the position the A-Mab case study adopts for its attribute assessment (CMC Biotech Working Group 2009).

Viral safety is evaluated modularly across the process, and clearance is claimed only for the steps that were spiked and studied under ICH Q5A (International Council for Harmonisation 2023a). Harvest and clarification is not one of them. No log reduction is credited to this step anywhere in the package, and none is claimed here.

1.4 Related documents

The documents that govern or consume this report are listed in Table 1.1. RA-001 set the characterization scope, PCP-004 is the executed protocol that fixed the design and the acceptance criteria before any run, and PCMR-001 consolidates this report with the other seven step reports for the transfer described in PTP-001.

Table 1.1: Documents related to this report.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope

Document ID	Title	Relationship
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-004	Process Characterization Plan (Harvest and Clarification (Step 4))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

Centrifugation followed by depth filtration is the established primary recovery operation of the Novacyte humanized IgG1 platform. It has been used to manufacture three prior platform products (X-Mab, Y-Mab and Z-Mab) and to supply clinical material for A-Mab itself. The equipment train, the filter media grade and the operating principle have not changed across those products. This experience constitutes prior product knowledge and may be applied directly to A-Mab, because the separation is governed by particle size and density and not by any property that differs between these molecules.

Prior knowledge also sets the mechanistic expectation that the study was designed to test. Applied centrifugal field governs how completely cells and large debris sediment (Stokes settling under an applied field), so it determines the solids load that the depth filter train has to retain. Depth filter loading governs how much of the available filter capacity is consumed, so it determines how close the train runs to breakthrough. Turbidity of the clarified harvest is the observable that both mechanisms end at, which is why it is monitored in process and why it is also treated as a parameter of the step in its own right.

One mechanism could in principle link this step to a quality attribute. Hydrodynamic shear in the feed zone of a disk-stack centrifuge can damage protein, and shear induced damage would appear as an increase in high molecular weight species (HMW), which is measured by size exclusion chromatography (AMV-3011). Platform experience with the three prior products gives no indication of such an increase, and the study measured size variants across the step specifically to test it. Beyond that single mechanism there is no path from a harvest parameter to product quality, and the risk assessment scored the step accordingly.

2.2 Quality attributes in scope

No drug substance attribute is set at this step, and the register assigns all 10 of them to other unit operations. The attributes that are present in the harvested stream, and that pass through this step unchanged, are listed in Table 2.1 with their acceptance criteria and their assigned criticality.

Table 2.1: Quality attributes present in the harvested stream, with the acceptance criteria and criticality assigned in the drug substance register. None is set or modified at this step.

CQA	Category	Acceptance	Criticality	Tool #1
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4

The seven attributes fall into three groups. The glycosylation and charge variant attributes (afucosylation, galactosylation, high mannose and acidic charge variants) are formed inside the cell and in the culture fluid, and prior knowledge shows that their distribution is not modified by the platform purification train. High mannose carries the highest criticality of the seven, and it is established during culture and reported in PCR-003. Aggregate is a size variant that the process both forms and removes, and it is the one attribute with a plausible mechanism at this step. Host cell protein and residual DNA are process impurities generated during culture, delivered by this step, and cleared downstream.

Both process impurities are cleared by the chromatographic steps that follow. Host cell protein is reduced by Protein A capture (PCR-005), by cation exchange (PCR-007) and by anion exchange (PCR-008). Residual DNA is reduced by Protein A capture and by anion exchange. The contribution of harvest and clarification to both is the delivery of a consistent, well clarified feed, and nothing beyond that. Aggregate is raised slightly by the low-pH hold (PCR-006) and reduced by cation exchange, which is the principal aggregate polishing step of the process.

2.3 Risk-based prioritization and parameter selection

RA-001 assessed every parameter of the process before any characterization study was executed, and its output is the study type assigned to each one. The rows for this step are given in Table 2.2.

Table 2.2: Pre-characterization risk assessment of the harvest and clarification parameters, as recorded in RA-001.

Parameter	Prospective failure mode	Effect	Initial RPN	Assigned study	Priority
Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	Affects clarification/yield (no CQA impact)	96	univariate	Low
Depth filter load	Depth filter load outside 80-160 L/m2	Affects filter capacity/turbidity (no CQA impact)	96	univariate	Low
Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)	96	univariate	Low

All three parameters scored identically, and the reason is visible in the effect column. None of them has a path to a critical quality attribute, so each was scored at severity 4, the level RA-001 labels slight and defines as an unlikely impact on product quality with no supplemental testing. Initial occurrence was scored 4 (low), because these parameters are held by platform controls and a departure from the intended setting acts on performance alone. Initial detection was scored 6 (moderate), which RA-001 defines as detection downstream and before certificate of analysis testing, and not during the unit operation itself. The initial risk priority number is 96 for each parameter. The in-process turbidity measurement described in Section 10 detects the consequence of a departure at the step itself, which the pre-characterization score does not credit.

Those are pre-characterization scores, and the decision they feed is the study type and not a classification. The initial number is not read against the risk priority number of 72 above which this package classifies a parameter as critical, because that threshold applies to the residual risk left after a characterization study. RA-001 ranks each parameter on its potential impact on quality attributes and on its potential to interact with other parameters, and a parameter that carries neither is assigned a univariate study, which is what happened to all 3 parameters of this step. The classification of a parameter is an outcome of the study and is given in Section 9.

That assignment is the reason this report contains no designed experiment, and the campaign as a whole splits the same way. Of the 8 unit operations characterized, 6 carry a multivariate design and 2 do not (Table 13.3). The two exceptions are the two steps that neither form nor clear a quality attribute, which are this step and ultrafiltration with diafiltration (PCR-010).

The ranges were chosen against the same prior knowledge. Each characterization range opens the routine control range by a factor between 2 and 3, which is wide enough to include the settings a plant would reach under a controller fault or an equipment change, and wide enough to expose the failure modes in Table 2.2 if they exist. The centrifugation range extends downward far enough to leave the depth filter train with a solids load it would not normally see, and the loading range extends upward past the point at which platform experience predicts that turbidity breakthrough begins. Opening them further would have added knowledge space with no prospect of a quality finding.

3 Materials and methods

3.1 Scale-down model and its qualification

A scale-down laboratory system was qualified as a model of the commercial harvest and clarification operation, under SOP-1001. The centrifugation stage is represented by a laboratory disk-stack centrifuge operated at the same equivalent settling area per unit feed rate as the commercial machine, so that the applied centrifugal field and the residence time in the bowl are both matched, and the clarification stage uses the same depth filter media grade and the same sterile filter membrane as the commercial train, scaled on filter area per unit volume of centrate.

The scale-independent variables were operated at the commercial target values during the qualification runs, and these are the applied centrifugal field, the depth filter loading and the feed properties (cell density, viability and broth solids). The scale-dependent variables, which are feed rate, bowl volume and the number of solids discharges per batch, were set at small scale to reproduce commercial performance and not commercial magnitude. The qualification compared centrate turbidity, post-filter turbidity and product recovery against the commercial-scale record across replicate runs, and the qualification report is held under SOP-1001 and is not reproduced here.

The warrant this gives is narrower than the warrant a chromatographic scale-down model gives. The model reproduces the separation that the commercial equipment performs and the product recovery it achieves, but it does not reproduce the full shear history of a commercial disk stack, because feed zone geometry does not scale linearly with throughput. Any claim in this report that rests on shear is therefore supported by the size variant measurement across the step and by platform experience, and not by the model alone. Confirmation at commercial scale is part of Stage 2.

3.2 Operation

The step was operated as the commercial process will be operated, and the operating procedures are the commercial ones. Whole cell broth is transferred from the production bioreactor to the centrifuge feed tank at the end of culture, and it is fed continuously to the disk-stack centrifuge at the applied centrifugal field given in Table 4.1, with intermittent solids discharge (SOP-2005). The centrate is collected and passed through the depth filter train at the specified loading, then through a sterile filter into the clarified harvest hold vessel (SOP-2006), where it is held at controlled temperature until it is loaded onto Protein A capture.

Materials that are controlled to a platform specification were not characterized. The depth filter media grade, the sterile filter membrane and the flush and rinse buffers are identity and quality controlled on receipt (incoming inspection and certificate of analysis review), so their acceptance is a material control and not a process parameter, and a departure from it is handled as a material deviation. One of the two deviations recorded during execution concerns a depth filter lot, and it is described in Section 13.1.

3.3 Analytical methods

Every method used in the study is validated, and the controlled documents are listed in Table 3.1.

Table 3.1: Controlled procedures and validated analytical methods applied to this step.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-2005	Harvest by Continuous Disk-Stack Centrifugation	SOP
SOP-2006	Clarification by Depth and Sterile Filtration	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3015	Turbidity by Nephelometry (NTU)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation

Turbidity by nephelometry (AMV-3015) is the primary in-process method of the step. It reports the particulate content of the centrate and of the clarified harvest, and it is the method that detects a clarification failure. Its intermediate precision is 4 %RSD, with a limit of quantitation of 0.5 NTU that lies below the lower edge of the range studied. Host cell protein by ELISA (AMV-3012) and residual DNA by qPCR (AMV-3014) quantify the impurity burden delivered to capture. Size variants by SEC (AMV-3011) measure the aggregate content of the bioreactor sample and of the clarified harvest, which is the paired comparison that tests the shear mechanism described in Section 2.1. Validated performance is given in Appendix A for the methods that carry a performance record, and in the individual validation reports for the remainder.

3.4 Sampling plan

Samples were taken at four points in every run. The bioreactor sample is the input reference and is drawn at the end of culture, the centrate sample is drawn after the centrifuge and reports the performance of the first stage alone, the post-filter sample is drawn after the depth filter train, and the clarified harvest sample is drawn after sterile filtration and is the material that loads capture.

Turbidity was measured on the centrate and on both filtered samples (AMV-3015). Host cell protein, residual DNA and size variants were measured on the bioreactor sample and on the clarified harvest sample, so that every quality result in this report is a paired comparison across the step and not a single end-point value read against a specification. Replicate determinations at the set-point condition provide the precision estimate against which the range runs are read.

3.5 Statistical methods

This characterization campaign uses two study types, and the division between them governs what may be claimed from each. A screening design identifies which parameters have an effect on a response. A response surface model fitted over the parameters that screening selects is the predictive model, and it is the basis on which a multivariate design space is defined. A screening fit is not used to predict, because it is near-saturated and carries no curvature.

Neither design applies here. RA-001 assigned all 3 parameters of this step to univariate study, so no screening design was run, no response surface model was fitted, and no design space is defined at this step. A univariate study supports a proven acceptable range (PAR) for one parameter at a time, with the others held at their set-points, but it does not resolve interactions between parameters and it cannot define a multivariate operating region. This report claims only what that design supports.

Results are reported as the observed value against the range studied and against the normal operating range, together with the paired comparison of each quality attribute across the step. A quality attribute is treated as unaffected when the paired comparison lies inside the replicate precision of its method. The precision of the turbidity and host cell protein methods is given in Table 13.2, and the precision of the SEC and qPCR methods is recorded in the AMV-3011 and AMV-3014 validation reports. Commercial-scale process capability is estimated for the drug substance as a whole and not per step, by Monte-Carlo simulation of 2,000 batches with every process parameter varying inside its normal operating range, from which a one-sided capability index is computed against each acceptance criterion (Section 8).

4 Study design

4.1 Factors, ranges and the knowledge space

The three parameters, their set-points, their normal operating ranges and the ranges over which they were characterized are given in Table 4.1. The classification column is an outcome of the study and is justified in Section 9.

Table 4.1: Harvest and clarification parameters, with set-point, normal operating range, characterization range, final classification and study type.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Centrifugation (rcf)	g	9,000	8000–10000	6000–12000	KPP	univariate
Depth filter load	L/m2	120	100–140	80–160	KPP	univariate
Post-clarification turbidity	NTU	5	1–10	1–20	GPP	univariate

The knowledge space of the step is the box that those three characterization ranges define. Two of its edges carry the argument. The low centrifugation edge at 6,000 g passes the largest solids load to the filter train, and the high depth filter loading edge at 160 L/m² leaves that train with the least capacity in reserve, so the two of them together define the worst case for clarification.

Turbidity appears in the table as a parameter, and it is also a result of the step. It is an attribute of the stream, but it is the attribute against which the clarified harvest is accepted for loading onto capture, so it is controlled as an input to the next operation and is carried in the parameter register for that reason. Section 5.4 reports it as the observable that the other two parameters act on, and Section 9 classifies it as the in-process control it becomes in routine manufacture.

4.2 Univariate assessment

Each parameter was varied across its characterization range with the other two held at their set-points, which is the design RA-001 assigned and PCP-004 approved. Set-point runs were replicated at the start and at the end of the study, and they provide the precision against which the range runs are read.

Two further conditions were run alongside the single-parameter scans, because the mechanism makes them the bounding cases for the filter train. The first combines the low edge of the centrifugation range with the high edge of the depth filter loading range, which is the worst case for clarification, since the centrifuge passes the largest solids load forward and the filter has the least capacity in reserve to retain it. The second combines the high edge of centrifugation with the low edge of loading, which is the condition that consumes the most filter area per batch. These two conditions bound the interaction that the univariate scans cannot resolve, and they are the only multi-parameter conditions the study contains.

The evidence base for the conclusions in this report is therefore the univariate scans, the two bounding conditions, the paired quality comparisons across the step, and platform experience with three prior products.

5 Results

5.1 Nominal performance and mass balance

The step performs as the platform predicts at its set-point condition. Table 5.1 gives the nominal batch.

Table 5.1: Nominal-batch performance of harvest and clarification at the set-point condition.

Metric	Nominal batch
Step yield (fraction of input product mass)	0.977
Product mass out (g)	64,071
Post-clarification turbidity (NTU)	5

Of the 65,596 g of product mass entering the step, the clarified harvest carries 64,071 g forward, so the loss is 1,525 g, or 2.3% of the input. Two mechanisms account for it, and in both the product is held up and not degraded: antibody is retained in the wet cell mass that the centrifuge discharges, and antibody is held in the wetted volume of the depth filter train and the sterile filter. Neither loss is recoverable at commercial scale without a wash step that would dilute the capture load.

Step yield was characterized as independent of all three parameters over the ranges studied. The commercial-scale yield model carries a mean of 95.0% with a coefficient of variation of 2.0%, and the nominal batch realized 97.7%. That is 1.4 standard deviations above the model mean, so the batch sits inside the batch-to-batch variation the model carries and is not a favourable draw. Harvest is the first step at which product mass is lost, so the cumulative yield of 97.7% at the end of the step is the step yield itself, and the position of the step in the full train is shown in Figure 5.1. It is a mid-range contributor to the total loss, and the largest single step loss is at Cation Exchange Chromatography (PCR-007).

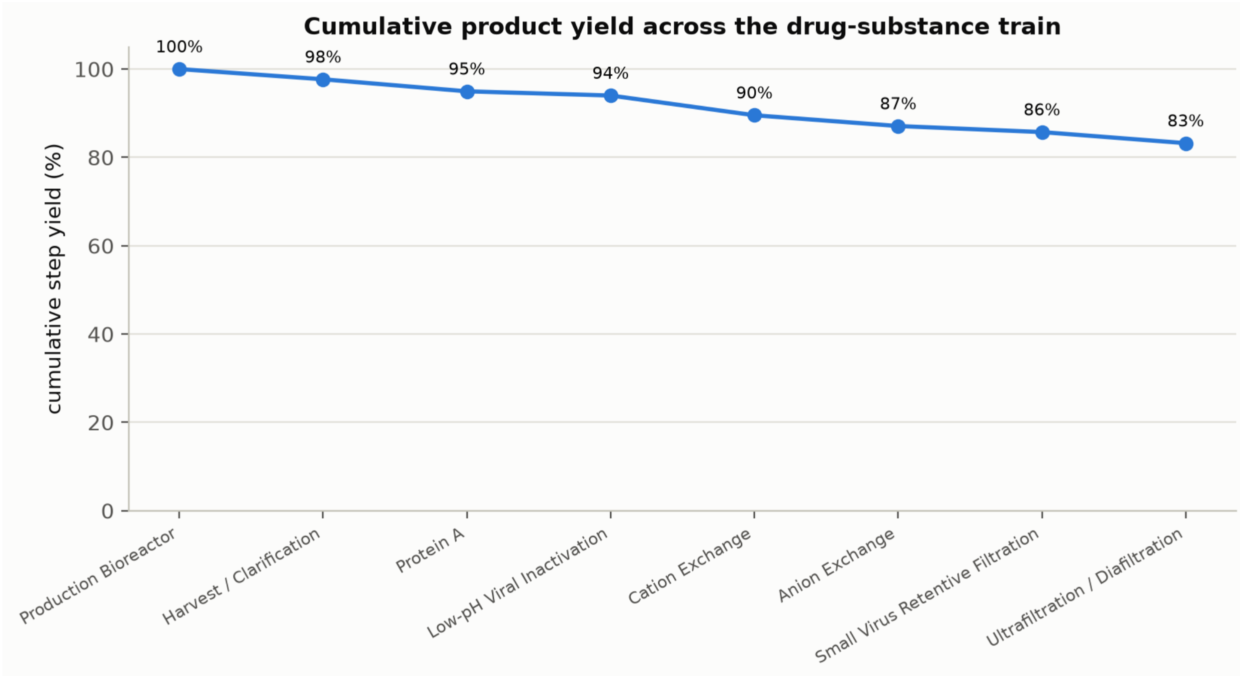

Figure 5.1: Cumulative product step-yield across the eight modelled unit operations of the A-Mab drug substance train.

5.2 Centrifugation across its characterization range

Applied centrifugal field governs the solids load passed to the depth filter train, and it has no effect on any product quality attribute. Over 6,000 to 12,000 g the step behaved as the sedimentation mechanism in Section 2.1 predicts, with less solid material carried forward at the higher settings, and the post-clarification turbidity stayed inside the range studied at every setting. The paired comparisons across the step for size variants (AMV-3011), host cell protein (AMV-3012) and residual DNA (AMV-3014) were within method precision at every setting.

The low edge of the range is the informative one. At 6,000 g the centrifuge passes the largest solids load forward, and the depth filter train has to retain it. That condition did not fail, it did not push post-clarification turbidity outside the range studied, and it is the condition that consumes filter capacity fastest, which is why it was combined with the high loading edge as one of the two bounding conditions of Section 4.2.

The high edge is uninformative by comparison. At 12,000 g the centrate is already essentially free of the particles that the depth filter is there to retain, so a further increase in the applied field buys nothing. Nor does it cost anything measurable. There was no increase in high molecular weight species at the high edge (SEC-HPLC, AMV-3011), which is the evidence that bears on the shear mechanism raised in Section 2.1. Shear in the feed zone of a commercial centrifuge does not scale from this model, so the finding is supportive and not conclusive, and it is confirmed at scale in Stage 2.

5.3 Depth filter loading and clarification capacity

Depth filter loading is the parameter that sets how close the train runs to breakthrough. Over 80 to 160 L/m² the filtrate turbidity rose with loading, in the direction platform experience predicts. A depth filter approaching the end of its capacity behaves that way, because retained solids reduce the available depth and progressively less of the challenge is captured. The same behaviour is seen with this media grade on the three prior platform products (X-Mab, Y-Mab and Z-Mab). In the univariate scan, with the centrifuge held at its set-point, post-clarification turbidity stayed inside 1 to 10 NTU across the whole loading range.

The bounding condition that combines the high loading edge with the low centrifugation edge produced the only post-clarification turbidity in the study that fell outside a normal operating range. Turbidity reached 12.5 NTU on that run, against a normal operating range that ends at 10 NTU. The value remains inside the characterization range, whose upper edge is 20 NTU. The excursion is 1.25 times the upper limit of the normal operating range, and it is far larger than the 4 %RSD intermediate precision of the turbidity method, so it is a real change in the stream and not analytical variation. The investigation, the root cause and the disposition are in Section 13.2.

Nothing about that excursion touches product quality. The paired comparisons of host cell protein, residual DNA and size variants across the step were within method precision on the affected run, as they were on every other run of the study. What the excursion establishes is the position of the normal operating range, and the argument for the upper loading limit of 140 L/m² is stronger for having a measured excursion above it than it would be from platform experience alone.

5.4 Post-clarification turbidity

Turbidity of the clarified harvest was held at 5 NTU at the set-point condition, and it stayed inside 1 to 20 NTU across every run of the study, including both bounding conditions. It is the only response of the step that is controlled in routine manufacture, and its value as a control comes from sitting downstream of both mechanisms. A centrifuge that under-performs and a depth filter train that is over-loaded both end at the same observable, so a single in-process measurement covers both failure modes and no separate test of centrate quality is required for batch release.

5.5 Impurity burden delivered to capture

The step delivers the process impurity burden generated during culture, and it removes none of it. Host cell protein in the clarified harvest is about 597,000 ng/mg at the culture set-point condition, and that burden is governed by the culture endpoint and not by anything that happens at this step. Extending culture duration from its set-point of 17 days to the upper edge of its characterized range at 19 days raises it by about 356,000 ng/mg, and it raises the residual DNA load by a factor of 1.7. Both increases follow from the culture endpoint, since a longer culture ends at a lower viability and releases more intracellular material into the broth. Culture duration is a parameter of the production bioreactor and its range is characterized in PCR-003.

None of the three parameters of this step changed either impurity measurably over the ranges studied. That is the expected result, because both impurities are soluble and both stages of this operation are particle separations. A depth filter train retains colloidal material and some lipid (by size exclusion and by adsorption onto the filter matrix), and it has no affinity mechanism for soluble protein or nucleic acid. The burden entering capture is controlled upstream, by holding the culture endpoint inside the ranges that PCR-003 characterizes, and it is reduced downstream by the three chromatographic steps named in Section 2.2.

5.6 Product quality across the step

No product quality attribute of A-Mab is altered by harvest and clarification. Every paired comparison across the step lay within the precision of its method, at the set-point condition and at both edges of all three parameter ranges, and the three attribute groups behave that way for three different reasons. The glycosylation and charge variant attributes cannot change here, because the step applies no chemical condition and no selective separation, and all four of them (high mannose, afucosylation, galactosylation and acidic charge variants) are established during culture and reported in PCR-003. Host cell protein and residual DNA cannot change here for the mechanistic reason given in Section 5.5. Aggregate is the only attribute with a mechanism at this step, and it is the one that was actually tested.

Size variants were measured on the bioreactor sample and on the clarified harvest of every run. No increase in high molecular weight species was detected across the step at any setting, including the high edge of the centrifugation range where hydrodynamic shear is greatest. The finding is bounded by the scale-down limitation in Section 3.1 and by the precision of the SEC method (AMV-3011). It is consistent with platform experience across the three prior products, it supports the conclusion that the step forms no size variant, and it does not by itself close the question at commercial scale.

This null result is retained in the knowledge space and is not discarded, because it is the evidence that the parameters of this step are robust with respect to product quality over ranges between 2 and 3 times the width of routine operation, and it is the basis on which all three are classified for process performance in Section 9.

6 Design space

No design space is defined for harvest and clarification. A design space is the multivariate region of the parameters within which the critical quality attributes remain inside their acceptance criteria (International Council for Harmonisation 2009), and it is defined at the steps that form or clear those attributes. This step does neither. There is no attribute for such a region to protect, no response surface model from which to construct one, and no interaction between the three parameters that a multivariate region would need to describe.

What the step contributes is an operating region defined on process performance. That region is the set of three normal operating ranges given in Table 4.1, nested inside the characterization ranges of the same table, and the relationship between the two is the usual one: the characterization ranges are the knowledge space, the normal operating ranges are the control space, and the proven acceptable ranges established in Section 7 lie between them. Movement of a parameter inside its normal operating range is routine operation. Movement outside a normal operating range but inside the proven acceptable range is handled as a deviation with a documented impact assessment, and the case that arose during this study is in Section 13.2. Movement outside a proven acceptable range is a change, and it is managed through the pharmaceutical quality system (International Council for Harmonisation 2008).

The worst case that the operating region must satisfy is the combination of the lowest applied centrifugal field with the highest depth filter loading (6,000 g and 160 L/m^2), because together they send the greatest solids challenge to the filter train. That corner was run as a bounding condition. It delivered clarified harvest at 12.5 NTU, which is inside the turbidity range studied and above the normal operating range, and it is the excursion investigated in Section 13.2.

Three bounds apply to everything in this section, and they apply again wherever this report states a result. The claim covers the ranges in Table 4.1 and no wider setting. It rests on a scale-down model that reproduces the separation and the recovery of the commercial equipment but not the shear history of a commercial disk stack. It assumes a feed drawn from a culture operated inside the endpoint ranges characterized in PCR-003, since the solids load and the impurity burden of the broth both depend on the culture endpoint. The design space of the A-Mab drug substance is defined at the steps that set and clear quality attributes, and it is consolidated in PCMR-001.

7 Proven acceptable ranges

7.1 Acceptance basis

The acceptance basis of a proven acceptable range at this step is process performance. At a step that governs a quality attribute, the acceptance criterion is the drug substance specification for that attribute, or, for a viral clearance attribute, the step contribution back-calculated from the cumulative requirement. Neither applies here, because no attribute is governed. A harvest and clarification parameter is proven against two criteria instead: clarified harvest delivered inside the turbidity range characterized at this step, which is 1 to 20 NTU, and step yield maintained.

That is a wider criterion than the one a routine batch meets. Post-clarification turbidity is held inside 1 to 10 NTU on every batch, and the clarified harvest is released to capture against that limit (Section 10). The two can differ here because no quality attribute depends on turbidity, so the routine limit is a process control and not a specification. A proven acceptable range that reaches past it records that the parameter did not fail anywhere it was tested, and it does not relax the release limit. At a step that governed an attribute the two criteria would be the same.

The two-analysis method used elsewhere in this package does not apply either. At a step with a fitted response surface model, each proven acceptable range is determined twice, once with the other parameters held fixed and once with them varying inside their normal operating ranges by Monte-Carlo of the fitted model, and both of those analyses require a model. No model was fitted at this step, so each range is established directly from the univariate scans, with the two bounding conditions of Section 4.2 standing in for the co-variation that the Monte-Carlo analysis would otherwise cover.

7.2 The ranges

Table 7.1 gives the proven acceptable range of each parameter with the basis on which it is proven.

Table 7.1: Proven acceptable ranges of the harvest and clarification parameters, with the performance basis on which each is proven.

Parameter	Unit	NOR	Char. range	Proven acceptable range	Acceptance basis
Centrifugation (rcf)	g	8000–10000	6000–12000	6000–12000	Clarified harvest delivered inside the turbidity range studied; step yield maintained
Depth filter load	L/m2	100–140	80–160	80–160	Depth filter capacity sufficient for the batch volume; clarified harvest delivered inside the turbidity range studied
Post-clarification turbidity	NTU	1–10	1–20	1–20	Feed acceptable to Protein A capture; in-process limit, no drug substance criterion

For all 3 parameters the proven acceptable range is the full characterization range, and the normal operating range sits inside it with margin on both sides. At the low centrifugation edge the binding consideration is filter capacity and not product quality, and at the high depth filter loading edge it is post-clarification turbidity, which reached 12.5 NTU when that edge was run together with the low centrifugation edge (Section 13.2).

Each range in Table 7.1 is proven over the interval given there and nowhere else, and it carries no information about settings beyond that interval. These are also univariate ranges, determined one parameter at a time with the others at their set-points, so they are not a multivariate region and the only evidence about combinations is the two bounding conditions. Since no attribute is governed here, that limitation costs the control strategy nothing. The binding constraint on the A-Mab operating region is the multivariate design space at the steps that form and clear quality attributes, and not any univariate range in Table 7.1.

8 Process capability and robustness

8.1 What is and is not attributed to this step

No commercial-scale process capability is attributed to harvest and clarification. A capability index compares the distribution of an attribute against its acceptance criterion, and this step sets no attribute for which such a comparison exists, so reporting one for the step would credit it with control that it does not exercise.

The three attributes that pass through the step in a form that could in principle be affected (host cell protein, residual DNA and aggregate) do have a drug substance capability, and it is given in Table 8.1. The estimate is a Monte-Carlo simulation of 2,000 commercial-scale batches with every process parameter varying inside its normal operating range, so it carries the variation of the whole train and not the variation of this step alone.

Table 8.1: Commercial-scale capability of the attributes carried through this step, from Monte-Carlo simulation of the whole process.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Aggregates (HMW)	H	upper	0–5	0.753 ± 0.088	16.1
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.14
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8

Host cell protein is the tightest of the three, at a capability index of 6.14. Its simulated mean is 15.2 ng/mg against an upper limit of 100 ng/mg, so the margin to the limit is 84.8 ng/mg. Residual DNA reaches 10.81 and aggregate 16.11. No acceptance threshold for a capability index is defined in this package, so each attribute is read against its own limit and its simulated mean in Table 8.1. Every one of those results is earned downstream, by the steps named in Section 2.2 for the two impurities and by cation exchange (PCR-007) for aggregate. The tightest capability anywhere in the drug substance is 1.51, at Viral clearance — MVM (parvovirus), which is set at anion exchange (PCR-008) and is unrelated to this step.

8.2 Robustness of the step itself

The robustness of harvest and clarification is expressed in the two responses that it does control. Step yield is characterized as independent of all three parameters across their ranges, with a commercial-scale mean of 95.0% and a coefficient of variation of 2.0%, and post-clarification turbidity stayed inside 1 to 20 NTU across every run, including the two bounding conditions, one of which is the excursion run of Section 13.2.

The three bounds stated in Section 6 apply to that robustness claim without change, and the feed assumption carries most of the weight. A broth with a lower final viability carries a higher solids and impurity load into this step, so the robustness holds for a feed drawn from a culture operated inside the endpoint ranges of PCR-003. Confirmation at commercial scale is a Stage 2 activity and is not claimed here.

9 Parameter classification

Each parameter was classified under SOP-4001, from the effects observed in the study and from the risk of operating outside the characterized region. The classifications are in the final column of Table 4.1, and the reasoning is as follows.

- Centrifugation, the applied centrifugal field, is a key process parameter. It governs the solids load that the depth filter train carries, as the study confirmed over 6,000 to 12,000 g, and it showed no effect on any quality attribute anywhere in that range. It is monitored and held inside 8,000 to 10,000 g because process consistency depends on it, which is what the key classification is for.
- Depth filter load is a key process parameter, on the same basis and with one additional piece of evidence. It acts directly on post-clarification turbidity, it produced the only excursion of the study when its upper edge was combined with the low centrifugation edge, and it showed no effect on any quality attribute. Its normal operating range ends at 140 L/m^2, below the loading at which that excursion was recorded.
- Post-clarification turbidity is a general process parameter. It is the acceptance condition that the clarified harvest meets before capture and it is measured on every batch, but no quality attribute of the drug substance depends on its value inside 1 to 20 NTU. It is controlled as an in-process limit and treated as a monitored attribute of the stream, not as a quality-linked parameter.

No parameter of this step is a critical process parameter or a well controlled critical process parameter. That outcome follows directly from the register: no critical quality attribute is set here, and no parameter of the step has a demonstrated effect on one. Both of the classes that are used (2 key process parameters and 1 general process parameter) are still carried into the control strategy, because their monitoring and control keep the process consistent and predictable.

The null results behind those classifications are retained in the knowledge space and are not dropped, since they are the evidence that allows the step to be operated to process performance limits alone, and they are what a future change to the equipment or to the media grade would be assessed against.

10 Contribution to the control strategy

Harvest and clarification contributes four elements to the A-Mab control strategy, and it contributes nothing to product quality directly. They are consolidated across the process in PCMR-001.

The first element is parameter control. Applied centrifugal field is held inside 8,000 to 10,000 g and depth filter loading inside 100 to 140 L/m², both under the operating procedures for the step (SOP-2005 and SOP-2006). Both are key process parameters, so both are monitored for process consistency and neither carries a quality-linked control limit.

The second element is in-process monitoring. Post-clarification turbidity is measured on every batch by nephelometry (AMV-3015) and is required to lie inside 1 to 10 NTU before the clarified harvest is released to capture. This single measurement covers both failure modes of the step, for the reason given in Section 5.4, and it is the point at which a clarification problem is caught before it reaches the capture column.

The third element is material and life-cycle control. Depth filter media lots are qualified on receipt and are used within their expiry, the pre-use flush volume specified in SOP-2006 is confirmed before every use (a requirement tightened as a direct result of Section 13.1), and centrifuge maintenance and the disk-stack inspection interval are governed by SOP-2005.

The fourth element is the feed that the step delivers. The clarified harvest is the defined input to Protein A capture and its acceptance is the turbidity limit above, but the size of the impurity burden it carries is not controlled here at all, so PCR-005 characterizes the capture step against the feed it actually receives.

The step does not reduce host cell protein or residual DNA toward the drug substance limits, and that reduction is delivered by Protein A capture, cation exchange and anion exchange. It makes no viral clearance claim (International Council for Harmonisation 2023a).

11 Discussion

The step is now characterized to the depth its risk warrants, and the outcome is the one RA-001 anticipated: two parameters showed the expected effect on clarification performance, none of the three touched a quality attribute, and the step is left controlled for process consistency and monitored by a single in-process test.

Confidence in the scale-down model is high for what the model was used to conclude, and it is lower for one specific question. Separation performance, filter capacity and product recovery all scale on well established principles (equivalent settling area, filter area per unit volume), and the model matched the commercial-scale record on each of them during qualification. Hydrodynamic shear in the feed zone of a disk-stack centrifuge does not scale in the same way. The report therefore does not rest the aggregate conclusion on the model alone, and instead rests it on the paired size variant comparisons across the step, on platform experience with three prior products, and on confirmation at commercial scale during Stage 2.

Two deviations occurred during execution, and the more consequential one was a turbidity excursion to 12.5 NTU on the bounding run that combined the top of the depth filter loading range with the low edge of the centrifugation range, against a normal operating limit of 10 NTU. The value stayed inside the characterization range, the paired quality comparisons on that run were unaffected, and the run was retained. What the excursion changed is the confidence in the loading limit, which is now supported by a measured excursion above it. The second deviation concerned the pre-use flush of a depth filter lot (LOT-DF-4408) and led to a tightened flush requirement in SOP-2006. Neither deviation altered a classification, a range or a reported result.

The conclusions of this report are bounded in three ways, and the first is the shear question above. A univariate design cannot resolve interactions, so the only evidence about parameter combinations is the two bounding conditions of Section 4.2. That is acceptable because no quality attribute is at stake here, and it would not be acceptable at a step that governs one. The ranges were also not confirmed at commercial scale at their edges, which is general to Stage 1 and is addressed by the PPQ campaign.

No parameter of this step required the most stringent form of control, and the reason is the absence of a quality linkage and not the presence of a wide margin. If a future change gave this step such a linkage, for example a change of filter media chemistry that introduced an extractable, the classification would have to be revisited and the risk assessment reopened.

12 Conclusions

Harvest and clarification is well understood over the ranges characterized, and it is ready to support Stage 2 at the ranges reported.

The step has 3 parameters, all characterized univariately across ranges between 2 and 3 times the width of routine operation, and the classification is 2 key process parameters and 1 general process parameter. None is quality-linked, and the proven acceptable range of each is its full characterization range (Table 7.1).

The step forms no critical quality attribute and modifies none of the 7 attributes present in the harvested stream. It delivers the process impurity burden generated during culture to Protein A capture and removes none of it, so its contribution to the impurity attributes is the delivery of a consistently clarified feed. At commercial scale those attributes meet their acceptance criteria with wide margin, the tightest being host cell protein at a capability index of 6.14, and that margin is earned by the clearance steps reported in PCR-005, PCR-007 and PCR-008. No viral clearance is claimed for this step.

The 2 recorded deviations were investigated and dispositioned as retained, and neither changed a classification, a range or a result. Every conclusion above carries the three bounds of Section 6, and this report rolls up into PCMR-001.

13 Deviations from the plan

Execution of PCP-004 recorded 2 deviations. Each was investigated, each was assessed for impact on the study and on the material, and each was dispositioned as retained. No run was excluded, no result was recalculated, and no range or classification in this report depends on a deviated condition. The register is given in Table 13.1.

Table 13.1: Deviations recorded during execution of PCP-004.

Deviation	Summary	Detected during	Disposition
DEV-004-01	Depth-filter lot pre-use flush turbidity above alert	pre-use filter flush	retained
DEV-004-02	Post-clarification turbidity excursion above NOR on one run	in-process turbidity monitoring	retained

The first entry was detected before product contact and concerns a material. The second was detected in process and concerns the product stream.

13.1 DEV-004-01, depth filter pre-use flush

The pre-use flush of depth filter lot LOT-DF-4408 gave a filtrate turbidity of 6 NTU, above the alert level applied at pre-use flush under SOP-2006. A depth filter that sheds media above the alert level at flush can carry that material into the product stream, and the finding was raised before the lot was used.

The investigation reviewed the flush records for the lot and compared the delivered flush volume against the volume that the procedure specifies. The delivered volume was short of the specified volume, and the root cause was recorded as an insufficient initial flush volume. The lot was inside its expiry date (2027-01-31) and had passed its incoming identity and quality checks, so no material disposition question arose beyond the flush itself. The flush was extended and repeated, the filtrate turbidity fell below the alert level, and the lot was released for use.

The impact assessment rests on when the measurement is taken. Pre-use flush turbidity is measured on flush water before any product contact, so it reports on the condition of the filter media and not on the product stream. The run that used LOT-DF-4408 delivered clarified harvest inside the characterization range for turbidity, and its paired quality comparisons (host cell protein, residual DNA and size variants) were within method precision, so the deviation was dispositioned as retained and the run contributes to this report unmodified. The corrective action is the flush volume confirmation now required before every use under SOP-2006, which is carried into the control strategy in Section 10.

13.2 DEV-004-02, post-clarification turbidity excursion

Post-clarification turbidity reached 12.5 NTU on one run, against a normal operating range that ends at 10 NTU. The excursion was detected in process by turbidity monitoring under AMV-3015, at the point at which the clarified harvest is tested before release to capture. It is the only value in the study that fell outside a normal operating range.

The investigation examined the run conditions and the analytical result in turn. The run was the bounding condition that combines the top of the loading range with the low edge of the centrifugation range, which Section 4.2 identifies as the worst case for clarification, and the root cause was recorded as depth filter loading near its maximum. The loading itself was inside the range that PCP-004 approved for study, so the departure is in the response and not in the setting. The analytical result was then confirmed against the method: the intermediate precision of AMV-3015 is 4 %RSD, so a value 1.25 times the upper limit of the normal operating range cannot be explained by analytical variation. The excursion is real.

The excursion lies inside the characterization range for turbidity, whose upper edge is 20 NTU, so the run was executed inside the knowledge space of the step and inside the proven acceptable range of Table 7.1. No critical quality attribute depends on post-clarification turbidity, and the paired comparisons of host cell protein, residual DNA and size variants on that run were within method precision. The deviation was dispositioned as retained, and the run is included in this report.

The excursion also earns its place in the control strategy. It is direct evidence that the upper limit of the depth filter loading range at 140 L/m^2 is set where it needs to be, and it is the reason the turbidity limit of 10 NTU is retained as a release condition for the clarified harvest and not as a trend alert.

Appendix A — Analytical methods

The validated performance of the methods that carry a performance record is given in Table 13.2. The performance of the remaining methods listed in Table 3.1 is recorded in their individual validation reports and is not reproduced here.

Table 13.2: Validated performance of the analytical methods applied at this step.

Method	Title	Precision (%RSD)	LOQ	Accuracy (%)	Variance share
AMV-3015	Turbidity by Nephelometry (NTU)	4	0.5 NTU	98	0.38
AMV-3012	Host-Cell Protein by ELISA	9.5	1 ng/mg	95	0.4

Appendix B — Characterization scope of the campaign

Table 13.3 gives the characterization scope of every unit operation with the plan and report that cover it. It is reproduced here because it places the univariate design of this study against the six steps that carry a multivariate design.

Table 13.3: Characterization scope by unit operation, with the number of parameters assigned to multivariate and to univariate study.

Step	Unit operation	Parameters	Multivariate	Univariate	Documents
3	Production Bioreactor	9	5	4	PCP-003 / PCR-003
4	Harvest and Clarification	3	0	3	PCP-004 / PCR-004
5	Protein A Chromatography	6	4	2	PCP-005 / PCR-005
6	Low-pH Viral Inactivation	4	3	1	PCP-006 / PCR-006
7	Cation Exchange Chromatography	5	4	1	PCP-007 / PCR-007
8	Anion Exchange Chromatography	5	4	1	PCP-008 / PCR-008
9	Small-Virus Retentive Filtration	2	2	0	PCP-009 / PCR-009
10	Ultrafiltration / Diafiltration	3	0	3	PCP-010 / PCR-010

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